

Generative Paper 1: Infinite Generative Field as the Origin of Physical Reality

Author Information

Author: Mortaza Zabihi

Affiliation: Independent Researcher

Email: m.zabihi.research@gmail.com

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Abstract

Modern physics successfully explains a vast range of natural phenomena, yet every existing framework begins from structures that are already assumed to exist. Space, time, particles, fields, and geometry serve as the primitives upon which physical laws are formulated. While this approach has been remarkably successful, it leaves a more fundamental question unanswered:

What is the simplest possible starting point before any physical structure is assumed?

This paper proposes a function-first foundation for addressing this question. Instead of treating geometry or physical entities as primitive, it begins with a single recursively self-generating function. At this most fundamental level, no space, no time, and no external source of variation are assumed to exist. Consequently, every new state of the system must arise entirely from the function itself.

This requirement leads naturally to the recursive relation

$$F_{n+1} = F_n + F'_n$$

where the derivative is interpreted not as a rate of change with respect to an external coordinate, but as the function's intrinsic generative variation.

Within this perspective, recursive transitions provide the first and most primitive interpretation of time: at the foundational layer, temporal progression is identified with the ordered sequence of generative transitions rather than with an independently existing dimension.

The purpose of this paper is not to derive physical laws directly, but to establish the minimal mathematical foundation upon which progressively richer structures—including geometry, spacetime, and physical interactions—may later emerge. In doing so, it shifts the starting point of fundamental physics from **geometry-first** to **function-first**.

Keywords: Infinite Generative Field; IGF; Generative Architecture; Generative Framework;

1. Introduction

Every physical theory begins somewhere.

Classical mechanics begins with space, time, and particles. General relativity begins with a spacetime manifold. Quantum field theory begins with quantum fields defined over spacetime. Although these theories differ profoundly in their mathematical formulation, they share a common assumption: **they start from a universe that already possesses structure.**

For the purpose of describing physical phenomena, this is entirely reasonable. Physics seeks to explain how existing structures behave, interact, and evolve. It is therefore natural to assume a geometric background before constructing the corresponding physical laws.

A different question, however, lies beneath this approach:

What is the minimal assumption required before any physical structure can exist?

If space is assumed, geometry has already been introduced.

If time is assumed, the existence of ordered change has already been accepted.

If particles or fields are assumed, stable physical entities already exist.

None of these concepts therefore represents the true beginning of a foundational theory. Each already presupposes a level of organization whose own origin remains unexplained.

The motivation for this work originated from a different line of inquiry. Rather than asking how physical systems evolve, the original question was:

What is the one aspect of existence that cannot be denied?

This investigation initially led to the concept of **awareness**—not as a psychological phenomenon, but as the minimal condition under which anything can become defined. As the framework evolved, however, it became clear that awareness itself was not the deepest concept. The more fundamental idea was **generation**: the process through which an undefined state gives rise to progressively more defined states.

This realization shifted the focus of the research. The objective was no longer to construct a theory of awareness, but to identify the simplest mathematical object capable of generating increasingly rich structures without presupposing any physical background.

The central proposal of this paper is that such an object is a **recursively self-generating function.**

At the foundational layer of the Generative Architecture, nothing else is assumed. There is no space in which the function exists, no time through which it evolves, and no external mechanism that drives its behavior. The function is the only primitive object, and every subsequent structure must arise from its own intrinsic generative process.

This represents a reversal of the conventional order used in fundamental physics. Instead of beginning with geometry and describing functions within it, we begin with a function whose recursive organization may eventually give rise to geometry itself. The goal is therefore not to reinterpret existing physical theories, but to investigate whether a simpler mathematical foundation is possible.

The remainder of this paper develops the consequences of this single assumption. We first examine why a recursively evolving function necessarily requires an intrinsic source of variation. We then show how this requirement leads naturally to a simple recursive equation, from which the first interpretation of time emerges as the ordered sequence of generative transitions. Finally, we discuss how this function-first perspective provides the conceptual foundation for the subsequent layers of the Generative Architecture.

2. Why Physics Cannot Be the Starting Point

Every foundational theory must begin with a set of primitive assumptions. These primitives are accepted without explanation and provide the basis upon which all subsequent concepts are constructed.

The question addressed in this paper is therefore not **how physics works**, but **where physics should begin**.

At first glance, the answer seems obvious. One might choose space, time, particles, or fields as the fundamental ingredients of reality. Indeed, modern physics has achieved extraordinary success by adopting precisely these concepts as its starting point.

However, from a foundational perspective, each of these concepts already contains structure.

Space is not merely an empty background. It already assumes extension, separation, and geometric relations.

Time is not simply a parameter. It already introduces an ordered sequence in which one state is distinguished from another.

Particles are already stable entities possessing identity.

Fields already describe structured quantities distributed over an underlying domain.

These concepts are therefore not primitive. They are organized structures whose own existence still requires explanation.

The difficulty can be summarized as a simple principle:

A concept cannot be considered fundamental if it already presupposes the structure it is intended to explain.

This observation motivates a different strategy. Rather than beginning with physical objects, we seek a starting point that contains **no physical assumptions whatsoever**.

At the foundational layer, we therefore assume:

- No space.
- No time.
- No geometry.
- No particles.
- No fields.
- No physical laws.

This is not because these concepts are unimportant, but because each already represents a stage of organization that should emerge from something more primitive.

Removing these assumptions leaves a natural question:

What is the simplest mathematical object capable of existing without any physical background?

Whatever this object is, it must satisfy only one essential requirement: it must be capable of generating successive states without relying on anything external to itself.

Since no external space exists, nothing can act from outside.

Since no external time exists, no independent parameter can drive its evolution.

Since no other primitive objects exist, no external interaction can generate change.

The origin of every new state must therefore lie entirely within the object itself.

This requirement immediately rules out many familiar mathematical constructions. Most mathematical objects are defined relative to an already existing framework—a coordinate system, a geometric space, or an independent variable. Such structures are unavailable at the foundational layer because they are precisely what we seek to explain.

The search therefore narrows to the simplest possible object capable of **self-generation**.

In the next section, we argue that this object is a **recursively self-generating function**. Unlike physical entities, it requires no geometric interpretation, no external coordinates, and no predefined background. It provides the minimal mathematical primitive from which increasingly rich levels of organization may be constructed.

3. The Minimal Primitive

Having removed every conventional physical assumption, we now face the central question of this paper:

What is the simplest mathematical object that can exist without presupposing space, time, geometry, or physical law?

The answer should be as minimal as possible. Every additional primitive increases the complexity of the theory and therefore requires its own explanation. If the objective is to construct a truly foundational framework, the starting point must contain only what is strictly necessary.

The only essential requirement is that the primitive must be capable of **generation**. A completely static object cannot produce new structure. If nothing can change, then nothing can emerge.

We therefore introduce a single primitive object, denoted by F .

At this stage, no interpretation is attached to (F) . It is not a physical field, a particle, a wavefunction, or a geometric object. It is simply a **generative function**.

Likewise, the word *function* should not be understood in its conventional physical sense. Normally, a function is defined over an already existing domain such as space or time. Here, no such background exists. The function is introduced before any geometric or physical interpretation has been established.

The defining property of (F) is that it is **recursive**. Rather than representing a fixed object, it continuously generates successive versions of itself,

$$F_0, F_1, F_2, \dots, F_n$$

where each term denotes a new generative state.

The index (n) should not be interpreted as physical time. At this stage, time has not yet been introduced. The sequence merely represents the logical order of recursive generation: one state is produced from another without assuming an external temporal dimension.

This distinction is essential.

In conventional mathematics, a sequence evolves **in** time or with respect to an independent parameter. In the present framework, the sequence itself is more fundamental than time. It is simply an ordered chain of generative states, leaving open the question of whether time might later emerge from this ordering.

The next question naturally follows:

How does one recursive state become the next?

The most general recursive relation can be written as

$$F_{n+1} = F_n + \Delta$$

where (Δ) represents the generative transition responsible for producing the next state.

At this point, nothing has been assumed about the nature of (Δ). It merely denotes whatever distinguishes one recursive state from the next.

However, a crucial constraint already exists.

Since (F) is the **only** primitive object in the framework, the source of (Δ) cannot lie outside the function. There is no external space, no external time, no surrounding environment, and no second object capable of introducing variation.

The transition must therefore be generated entirely by the function itself.

This conclusion follows directly from the principle of minimality. Once a single primitive object has been adopted, every aspect of its evolution must originate internally. Otherwise, an additional primitive would have been introduced.

The next section investigates the consequences of this requirement and shows that the simplest consistent realization is obtained when the generative transition is identified with the function's own intrinsic variation, leading naturally to the recursive equation that forms the foundation of the Generative Architecture.

4. Intrinsic Variation

The previous section established the existence of a recursively self-generating function,

$$F_0, F_1, F_2$$

The next question is immediate:

What allows one recursive state to become the next?

A recursive system cannot evolve unless each new state differs from its predecessor. If every state were identical,

$$F_{n+1} = F_n$$

then recursion would still exist as a formal sequence, but nothing would actually be generated. The system would remain completely static.

Recursive generation therefore requires **variation**.

Without assuming anything about its origin, the most general recursive relation can be written as

$$F_{n+1} = F_n + \Delta$$

where (Δ) denotes the transition that transforms the current state into the next.

The problem is that the origin of (Δ) has not yet been explained.

In conventional physics, variation is usually attributed to external causes. A system changes because it interacts with its surroundings, evolves in time, or responds to external forces. These explanations are possible because an external framework already exists.

At the foundational layer considered here, no such framework is available.

There is no surrounding space.

There is no independent time parameter.

There is no second object capable of interacting with the function.

The function itself is the only primitive object in the theory.

Consequently, the transition responsible for recursive evolution **cannot originate from outside the function**.

This leaves only one logically consistent possibility:

The variation must be generated by the function itself.

The simplest mathematical realization of this idea is to identify the generative transition with the function's own derivative,

$$\Delta = F'_n$$

Substituting this expression into the recursive equation gives

$$F_{n+1} = F_n + F'_n$$

This equation is the central result of the present paper.

Its importance does not lie in its algebraic simplicity, but in the interpretation it provides. The derivative is no longer viewed merely as a rate of change with respect to an external coordinate. Instead, it represents the function's own **intrinsic generative variation**—the internal mechanism through which one recursive state produces the next.

This interpretation differs fundamentally from the conventional role of differentiation.

In classical mathematics,

$$\frac{dF}{dx}$$

measures how a function changes as an independently existing variable (x) changes.

Here, no such variable has yet been introduced.

There is no space.

There is no time.

There is no external parameter.

The derivative therefore cannot be understood as measuring change **with respect to** something outside the function. Instead, it represents the minimal internal distinction required for recursive generation itself.

The equation

$$F_{n+1} = F_n + F'_n$$

thus expresses the simplest self-generating process possible. Every new state is produced entirely from information already contained within the current state. No external driver, background geometry, or independent evolution parameter is required.

4.1 The First Interpretation of Time

The recursive equation also provides the first interpretation of time within the Generative Architecture.

At this foundational layer, time has not been assumed as an independent dimension. The only primitive process is the transition from one recursive state to the next.

Consider the limiting case

$$F'_n = 0$$

The recursive equation immediately becomes

$$F_{n+1} = F_n$$

No new state is generated.

No transition occurs.

Nothing progresses.

This observation suggests that **time does not exist independently of recursive generation**. If no generative transition occurs, there is no meaningful distinction between successive states.

Conversely, whenever the intrinsic variation is nonzero,

$$F'_n \neq 0$$

a new generative state is produced. The ordered succession of these transitions provides the simplest possible notion of temporal progression.

At the foundational layer of the framework, **time is therefore identified with the ordered sequence of generative transitions.**

It is important to emphasize that this is **not** proposed as a universal definition of time. It is only the first interpretation, valid at the most primitive level of the Generative Architecture.

As progressively richer organizational structures emerge, new notions of time may also emerge. The physical time measured by clocks, the relativistic time of spacetime geometry, and the temporal experience of observers belong to later architectural layers built upon this more primitive generative foundation.

Thus, within the present work, time is not treated as a background in which generation occurs. Rather, **generation itself provides the first condition under which time becomes meaningful.**

5. Function-First Mathematics

The recursive equation

$$F_{n+1} = F_n + F'_n$$

is intentionally simple. Its purpose is not to provide a complete physical theory, but to establish a different starting point for one.

The central idea of this paper can be summarized in a single shift of perspective:

Instead of beginning with geometry and defining functions upon it, we begin with a function from which geometry may eventually emerge.

This is what we call the **function-first** approach.

In conventional mathematics and physics, functions are almost always introduced within an already existing framework. A function is defined on a domain, derivatives are taken with respect to independent variables, and physical evolution is described relative to coordinates such as space or time.

The underlying order is therefore

$$\text{Geometry} \longrightarrow \text{Function} \longrightarrow \text{Dynamics}$$

Geometry comes first.

Functions are defined within that geometry.

Dynamics then describe how those functions evolve.

The present framework proposes the opposite order.

Function → Generation → Geometry

The function is the primitive object.

Recursive generation is its primitive operation.

Geometry is not assumed; it is expected to emerge only after sufficient generative organization has developed.

This reversal also changes the meaning of differentiation.

In classical calculus, the derivative measures change relative to an independent coordinate,

$$\frac{dF}{dx}$$

The coordinate (x) already exists before the derivative is defined.

Within the present framework, no such coordinate has yet been introduced. Consequently, the derivative cannot be interpreted as a geometric rate of change.

Instead, it represents the function's **intrinsic generative variation**—the internal distinction that allows one recursive state to generate the next.

The difference between the two viewpoints is summarized below.

Conventional Foundation

Space and geometry are assumed.

Functions are defined on a pre-existing domain.

Derivatives measure change with respect to external variables.

Time is an independent parameter.

Function-First Foundation

No physical background is assumed.

The function is the primitive object.

Derivatives represent intrinsic generative variation.

Time emerges from recursive transitions.

The purpose of this reinterpretation is not to replace conventional mathematics.

Once geometry emerges, ordinary differential calculus remains the appropriate language for describing physical systems. The present framework instead asks a more primitive question:

What mathematical structure is required before geometry itself can exist?

The answer proposed in this paper is remarkably simple.

A single recursively self-generating function.

An intrinsic source of variation.

And the recursive equation

$$F_{n+1} = F_n + F'_n$$

Together, these provide the minimal mathematical foundation from which progressively richer structures may later be developed.

The following paper introduces the second foundational layer of the Generative Architecture by addressing an equally fundamental question:

What is the minimal condition under which such a generative function can become definable?

There, the concept of **vacuum** is reconsidered—not as empty space or the ground state of quantum fields, but as the minimal condition of definability itself.

6. Conclusion and Outlook

This paper has explored a single foundational question:

What is the simplest possible starting point for physics?

Rather than beginning with space, time, geometry, particles, or fields, we sought a starting point that does not presuppose any physical structure. The goal was not to construct a new physical theory directly, but to identify the minimal mathematical primitive from which increasingly rich structures might eventually emerge.

The central proposal is that this primitive is a **recursively self-generating function**.

Because no external background exists at the foundational layer, every new state must arise from the function itself. This requirement leads naturally to the recursive equation

$$F_{n+1} = F_n + F'_n$$

where the derivative is interpreted as the function's **intrinsic generative variation** rather than as a rate of change with respect to an external coordinate.

This reinterpretation also provides the first notion of time within the Generative Architecture. At the foundational layer, time is not assumed as an independent dimension. Instead, it is identified with the ordered sequence of generative transitions. If no transition occurs, no temporal progression exists. Time, in this sense, is a consequence of generation rather than its prerequisite.

The perspective developed throughout this paper may therefore be summarized by three principles:

- **The primitive object is a recursively self-generating function.**
- **Generation is driven by intrinsic variation rather than external causes.**

- **Time emerges from recursive transitions instead of existing as a fundamental background.**

Together, these principles define the **function-first** foundation of the Generative Architecture. They do not replace existing physical theories; rather, they propose a level of description that precedes the assumptions upon which those theories are built.

This paper intentionally stops at this foundational layer. It does not attempt to derive geometry, spacetime, or physical interactions. Those require additional organizational principles that have not yet been introduced.

The next paper addresses the first of these questions:

If a generative function is the primitive object, what is the minimal condition under which it can become definable?

To answer this, the concept of **vacuum** is reexamined from a foundational perspective. Instead of treating the vacuum as empty space or the lowest-energy state of quantum fields, it is introduced as the **minimal condition of definability**. Together, these first two papers establish the conceptual foundation of the Generative Architecture:

- **Paper 1:** The minimal primitive — a recursively self-generating function.
- **Paper 2:** The minimal condition of definability — the foundational vacuum.

The subsequent papers build upon these foundations, introducing progressively richer organizational structures while preserving the central principle established here: **physical reality is approached not by assuming structure, but by allowing structure to emerge through successive stages of generation.**

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This paper represents one step within the ongoing development of the **Generative Architecture** framework. Subsequent papers will progressively extend the concepts introduced here into a unified theoretical structure.